

Agentic Causal Macro Intelligence: Evidence Contracts and Regime-Routing Failure under Misspecification

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Abstract

Regime-aware decision systems can route predictions through specialized experts, yet a well-separated latent state does not establish a valid causal mechanism. We develop the Causal Macro Transmission Framework (CMTF) as an identification-aware interface between proposed economic mechanisms, versioned evidence, and expert eligibility. A reference router checks declared identification status, provenance, numerical validity, and evidence availability before assigning normalized expert weights or abstaining. We evaluate a separate synthetic diagnostic that isolates confounding and regime-observation misspecification. Four scenarios, each with 200 independent seeds, compare pooled regressions, hard and soft filtered routing, and an oracle-state baseline over 2,000 training and 1,000 test observations. With observed confounding and a correctly specified observation model, soft routing reduces mean squared structural-coefficient error from 1.025 for pooled adjusted estimation to 0.373. Reversing the observation-to-regime mapping at test time raises soft-routing error to 2.890, above the unchanged pooled error of 1.025. With hidden confounding, even oracle-state routing retains error of approximately 1.004. These results separate state uncertainty from causal identification: improved routing cannot remove an omitted common cause, and correct local estimates can be misapplied when state semantics change. The contribution is an executable evidence contract and a reproducible failure diagnostic, not an end-to-end causal discovery system or evidence of trading profitability.

1 Introduction

Economic narratives often specify a pathway from a macroeconomic development to an asset price. Turning such a narrative into a decision requires more than predicting a market regime. The question must specify what changes, which outcome is measured, the time horizon, the information available at the decision, and the assumptions under which the effect can be recovered from observations. A latent-state label by itself supplies none of these guarantees.

The Causal Macro Transmission Framework (CMTF) organizes this boundary between mechanism proposals and decision eligibility. It continues the author’s earlier conceptual note (Olapade, 2026), while narrowing the empirical contribution to an executable routing contract and a controlled synthetic diagnostic. A neural component may propose a mechanism or forecast; a symbolic record specifies its evidence status and applicability. The implementation tested here consumes declared records. It does not use a language model to establish identification, learn an economic causal graph, or verify the truth of an assumption.

We distinguish three questions. Is a causal effect identified under a declared model? Can the execution system infer the current regime reliably? Does a proposed action match the direction, horizon, and information vintage of its supporting evidence? Failure in one cannot generally be repaired by success in another. The synthetic study isolates the first two; the router implements a minimal contract for declared evidence. Neither is presented as a full deployed neuro-symbolic trading system.

Our main finding is a reversal: filtered routing improves coefficient recovery under the nominal observation mechanism, but becomes worse than pooled adjustment after the observation-to-regime relationship is reversed. A second finding is that oracle knowledge of the regime does not remove hidden-confounding bias. Together these motivate explicit failure states rather than a single undifferentiated confidence score.

2 Positioning and contribution

Mixture-of-experts systems combine specialist outputs using input-dependent weights (Jacobs et al., 1991). Weighting addresses which prediction to use; causal identification addresses whether the target effect is recoverable from the assumed graph and observed data. These are different operations. DoWhy explicitly separates modeling, identification, estimation, and refutation (Sharma and Kiciman, 2020). CMTF adopts this separation as an interface constraint rather than introducing a replacement identification calculus.

Selective prediction permits a system to decline an output when a selection criterion is not met (Geifman and El-Yaniv, 2019). Our router’s abstention is a deterministic contract outcome when no record passes the declared checks. It does not learn a calibrated risk–coverage trade-off. Here coverage means the fraction of cases receiving an output, and risk means the error on those retained cases; neither is measured on a real decision dataset in this study.

The original macro-to-asset framing supplies a use case, not evidence that a particular economic graph is correct. The present contribution comprises a precise record-to-routing interface, executable contract checks, and a four-scenario diagnostic with a known data-generating process. Standard regression and filtering are used to expose failure modes, not claimed as new estimators.

3 Evidence records and routing

3.1 A declared causal question

Let X be a treatment variable, Y an outcome, and C a set of pre-treatment context variables. A treatment is the variable whose value is hypothetically changed; a context variable is observed before that change. For values x_1, x_0 and context c , a possible query is

$$\tau(c) = \mathbb{E}[Y \mid \text{do}(X = x_1), C = c] - \mathbb{E}[Y \mid \text{do}(X = x_0), C = c]. \quad (1)$$

The scalar τ (tau) is a defined causal contrast. $\mathbb{E}$ denotes an expected value, or average under the specified distribution. The operator do denotes intervention: set X externally rather than merely select observations with a certain X . The vertical bar means conditioning. This definition does not establish that the contrast can be estimated from the available data.

An evidence record should specify the query, graph version, identification argument, estimator version, units, horizon, availability time, expiry, and unresolved assumptions. An execution-state estimate is kept distinct from the pre-treatment context used for identification. Conditioning on a variable caused by the treatment can change or bias the target; a convenient state feature must not automatically be used as a causal adjustment variable.

3.2 The implemented contract

The reference implementation stores, for each expert k , a unique identifier; direction $d_k \in \{-1, 0, 1\}$; score s_k ; response r_k ; nonnegative penalty p_k ; availability and expiry times; a declared identification flag; and a provenance string. Scores, responses, and penalties are finite scalar inputs. They are assumed to be on a compatible dimensionless scale in this demonstration. Real effects with different units or horizons require explicit conversion before aggregation.

At decision time t , the router rejects nonfinite values, invalid directions, negative penalties, missing provenance, records declared unidentified, and records unavailable or expired at t . A duplicate expert identifier raises an error. The identification flag is supplied by the caller; checking it does not prove identification. The richer graph, units, horizon, and diagnostic checks described above are an interface specification, not fully implemented validators.

For eligible expert set $\mathcal{K}_t$, the chosen logit and normalized weight are

$$\ell_k = s_k + d_k r_k - p_k, \quad (2)$$

$$w_k = \frac{\exp(\ell_k - m)}{\sum_{j \in \mathcal{K}_t} \exp(\ell_j - m)}, \quad m = \max_{j \in \mathcal{K}_t} \ell_j. \quad (3)$$

Here ℓ (ell) is a scalar routing score, w_k is a nonnegative weight, $\exp$ is the exponential function, and j indexes eligible experts. The set $\mathcal{K}_t$ contains only records that passed the contract. Subtracting the maximum m leaves the normalized ratios unchanged while preventing exponential overflow. The weights sum to one. If the eligible set is empty, the output is an abstention record with reasons; the denominator is never evaluated.

The direction term rewards alignment with the signed response. For example, with score 0.2, response -0.8 , and penalty 0.1, a long proposal has logit $0.2 - 0.8 - 0.1 = -0.7$, while a short proposal has logit $0.2 + 0.8 - 0.1 = 0.9$. With only these two eligible experts, their weights are approximately 0.168 and 0.832. A future-dated proposal is rejected even if its numerical score is very large. The example explains contract behavior; these numbers are not calibrated trading preferences.

The algorithm scans records once for validation, computes eligible logits, and normalizes them. A dictionary stores rejection reasons keyed by expert identifier, and another stores eligible weights. For K records, both time and storage grow linearly, written $O(K)$: doubling the number of records approximately doubles the work and retained entries. Portfolio constraints and order execution occur after routing and are outside this demonstration.

4 Synthetic diagnostic

4.1 Data-generating process

Each dataset contains 3,000 sequential observations: 2,000 training and 1,000 test. Let $S_t \in \{0, 1\}$ be a latent regime with transition matrix

$$A = \begin{pmatrix} 0.97 & 0.03 \\ 0.03 & 0.97 \end{pmatrix}. \quad (4)$$

A is a chosen 2×2 matrix: row j gives the probabilities of the next regime conditional on current regime j . It encodes persistence, not a fitted market model. The initial regime is sampled uniformly. Independent standard-normal random variables C_t, U_t, V_t, W_t generate

$$X_t = C_t + U_t, \quad (5)$$

$$Y_t = \theta_{S_t} X_t + 2C_t + V_t, \quad (6)$$

$$M_t = \mu_{S_t} + \sigma_M W_t. \quad (7)$$

The scalar C_t is a common cause of treatment X_t and outcome Y_t ; U_t and V_t are independent disturbances. M_t is a noisy regime observation, not an outcome or treatment. The chosen coefficient vector θ (theta) is $(1, -1)$ except in the inactive-second-regime scenario, where it is $(1, 0)$. The chosen emission-mean vector μ (mu) is $(-0.7, 0.7)$, and $\sigma_M = 1.3$ (sigma) is observation noise standard deviation. Each vector has two entries, one per regime. All observations are synthetic and use arbitrary units.

The four scenarios are observed confounding, hidden confounding, test-time emission reversal, and an inactive second regime. In the hidden-confounder case C_t exists but is unavailable to the estimators. In emission reversal, test observations use $-\mu_{S_t}$ while the filter retains nominal means. Structural coefficients do not change. This isolates a change in how the regime is observed from a change in the underlying outcome mechanism.

4.2 Estimation and filtering

The training regime labels are supplied as oracle information, meaning the simulator's true states are revealed to the estimator. We estimate a slope $\hat{\beta}_k$ separately within each training state k . With observed confounding, ordinary least squares regresses Y on an intercept, X , and C ; otherwise it uses an intercept and X . The hat denotes an estimate. Pooled baselines fit the analogous regression without splitting by state.

The filter receives only the sequential observations M_t and the fixed nominal transition and emission parameters. Let $q_{t,k}$ be its posterior probability of state k . Prediction and observation updating are

$$\tilde{q}_{t,k} = \sum_{j=0}^1 q_{t-1,j} A_{jk}, \quad (8)$$

$$q_{t,k} = \frac{\tilde{q}_{t,k} L_k(M_t)}{\sum_{j=0}^1 \tilde{q}_{t,j} L_j(M_t)}. \quad (9)$$

$\tilde{q}$ is the predicted probability before seeing the current observation; L_k is the normal observation likelihood in state k . The first step uses prior $(0.5, 0.5)$. Multiplication by likelihood favors states whose observation model better explains M_t , then division makes probabilities sum to one. Filtering runs over the training and test observation sequence, without access to test X, Y, C , or true S .

Table 1: Mean structural-coefficient MSE with Monte Carlo standard error in parentheses, across 200 independent datasets per scenario. These uncertainty measures describe simulation replication, not confidence in a real economic mechanism.

Method	Observed confounder	Hidden confounder	Emission reversal	Inactive regime
Pooled unadjusted	2.048 (.0358)	2.048 (.0358)	2.048 (.0358)	1.270 (.0180)
Pooled available adjustment	1.025 (.0047)	2.048 (.0358)	1.025 (.0047)	.257 (.0013)
Filtered hard routing	.503 (.0066)	1.497 (.0098)	3.501 (.0095)	.126 (.0017)
Filtered soft routing	.373 (.0041)	1.365 (.0097)	2.890 (.0084)	.094 (.0010)
Oracle-state routing	.001 (.0001)	1.004 (.0040)	.001 (.0001)	.001 (.0001)

Hard routing selects the slope of the most probable state. Soft routing averages the two slopes using posterior probabilities. Oracle routing selects the slope of the true test state:

$$\hat{\theta}_t^{\text{hard}} = \hat{\beta}_{\arg \max_k q_{t,k}}, \quad (10)$$

$$\hat{\theta}_t^{\text{soft}} = \sum_{k=0}^1 q_{t,k} \hat{\beta}_k, \quad \hat{\theta}_t^{\text{oracle}} = \hat{\beta}_{S_t}. \quad (11)$$

$\arg \max$ returns the state index with the largest probability. Superscripts name methods, not powers. These estimated scalars target the structural coefficient at each test time, rather than the realized outcome Y_t .

For test length $T = 1,000$, coefficient mean squared error is

$$\text{MSE} = T^{-1} \sum_{t \in \text{test}} (\hat{\theta}_t - \theta_{S_t})^2. \quad (12)$$

MSE averages squared coefficient-estimation errors; lower is better. It is not price-forecast loss or portfolio return. We use 200 independent seed values, $20260906 + r$ for replicate index $r = 0, \dots, 199$, reusing seeds across scenarios and methods for paired comparisons. Four scenarios times 200 datasets times five methods produce 4,000 method-level result rows, not 4,000 independent datasets.

5 Results

With observed confounding and the nominal observation mechanism, soft routing reduces mean MSE from 1.024985 for pooled adjustment to 0.373368. The reduction is 63.6% relative to that particular baseline, calculated as one minus the ratio of the two mean errors. Hard routing has error 0.503279. The paired hard-minus-soft difference is 0.129911 with Monte Carlo standard error 0.002849. Soft routing benefits from representing uncertainty rather than making a discrete state choice.

The benefit is conditional. Under emission reversal, soft-routing MSE is 2.890335 and hard-routing MSE is 3.500872, both worse than pooled adjustment at 1.024985. Oracle-state error remains approximately 0.000968 because structural coefficients and the training estimates are unchanged. The failure therefore arises from applying otherwise useful local estimates through an incorrect observation-to-mechanism mapping.

With hidden confounding, oracle-state routing has error 1.003865. Correct state selection cannot remove omitted-variable bias. Soft routing improves on the pooled unadjusted baseline but still has error 1.365132; a relative gain must not be mistaken for identification. In the inactive-second-regime case, soft-routing error is 0.093935 versus 0.256786 for pooled adjustment. The outcome metric remains coefficient error: no economic value of abstention or inactivity is inferred.

6 Understanding the failure mechanisms

6.1 Why oracle regimes do not repair confounding

Within either regime, $X = C + U$ and $Y = \theta_k X + 2C + V$. Independent standard-normal C, U, V give $\text{Var}(X) = 2$ and $\text{Cov}(X, C) = 1$. Variance measures spread and covariance measures joint variation. The population unadjusted

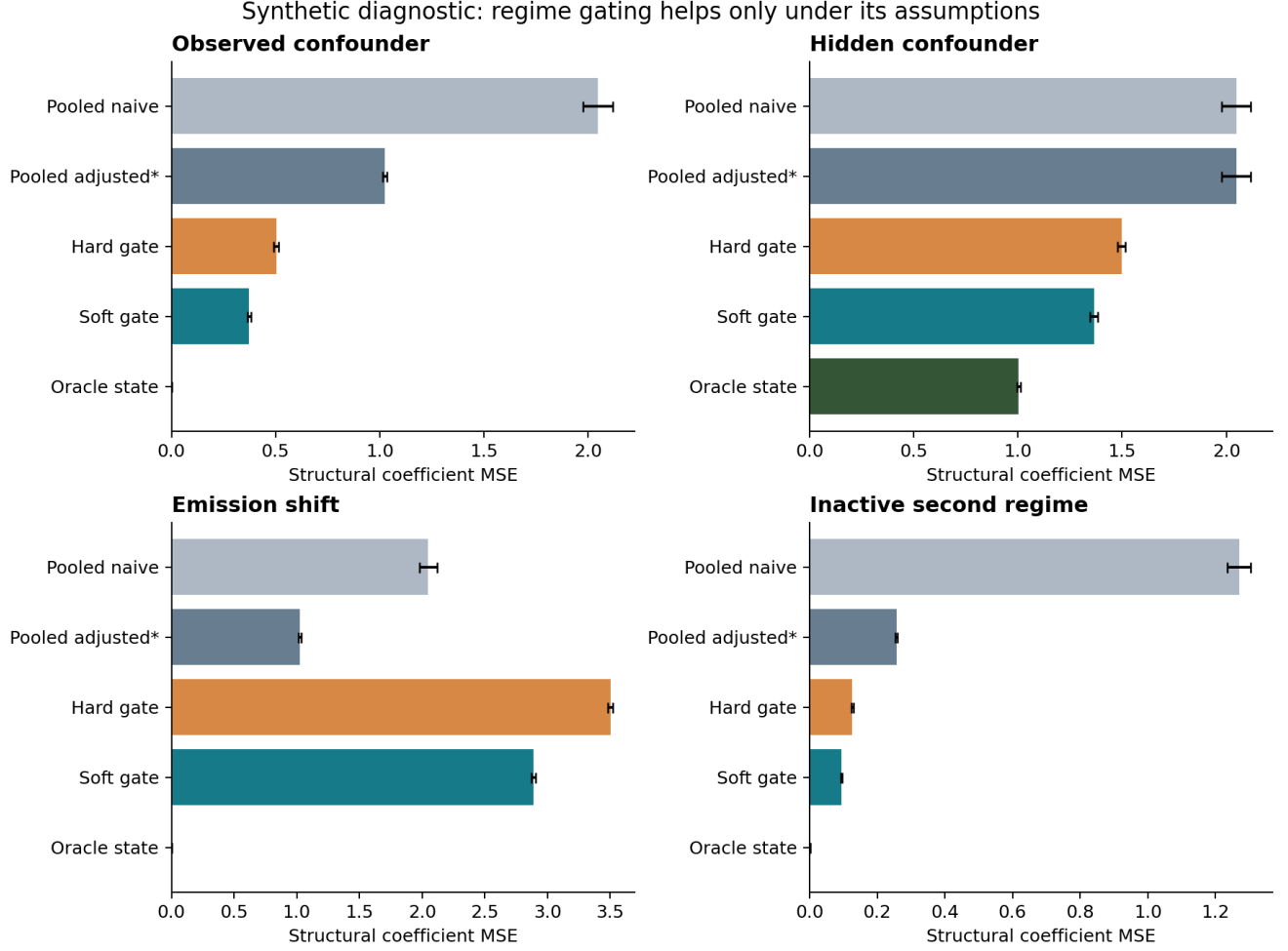


Figure 1: Routing performance depends on both confounding and the validity of the regime observation model. Error bars are 1.96 times the Monte Carlo standard error of mean MSE.

regression slope is therefore

$$\beta_k^{\text{unadjusted}} = \frac{\text{Cov}(X, Y \mid S = k)}{\text{Var}(X \mid S = k)} \quad (13)$$

$$= \theta_k + 2 \frac{\text{Cov}(X, C)}{\text{Var}(X)} = \theta_k + 1. \quad (14)$$

We first substitute the outcome equation into covariance, then use independence to remove covariance with V , and finally substitute the known variances. The coefficient bias is one, giving squared error one even with perfect state selection and unlimited training data. Finite-sample estimation explains the small departure of the simulated mean from one. This is a property of the declared synthetic model, not a theorem about all hidden-confounder settings.

6.2 Why probability weighting can help and still fail

Suppose for this explanation that the state coefficients are known exactly and the filter probabilities are correct. Let $B = \theta_{S_t}$ be the unknown coefficient and $\mathcal{I}_t$ the information available to the filter. For any proposed scalar estimate a , write $\bar{b} = \mathbb{E}[B \mid \mathcal{I}_t]$. Expanding the square gives

$$\mathbb{E}[(B - a)^2 \mid \mathcal{I}_t] = \text{Var}(B \mid \mathcal{I}_t) + (\bar{b} - a)^2. \quad (15)$$

The cross term vanishes because $B - \bar{b}$ has conditional mean zero. The first term does not depend on a , so choosing $a = \bar{b} = \sum_k q_{t,k} \theta_k$ minimizes conditional squared error. This familiar conditional-mean identity explains soft weighting under its assumptions; estimated coefficients and a misspecified filter need not attain that optimum.

For example, with state coefficients $(1, -1)$ and probabilities $(0.6, 0.4)$, soft weighting gives 0.2. Its conditional MSE is $0.6(1 - 0.2)^2 + 0.4(-1 - 0.2)^2 = 0.96$. Hard selection gives 1 and MSE $0.4(-1 - 1)^2 = 1.6$. If the true coefficient is 1 but a reversed observation model assigns probabilities $(0.1, 0.9)$, soft weighting gives -0.8 and squared error 3.24. A pooled coefficient near zero then has error near one. Reduced decisiveness helps under uncertainty; incorrect state semantics can dominate that benefit.

7 Reproducibility and limitations

All diagnostic data are generated from declared random seeds; no proprietary market data, model API, or accelerator is required. The artifact contains the generator, regressions, filter, router, tests, raw replicate results, and aggregation instructions. Regression design matrices have either two columns (intercept and treatment) or three (also confounder). Each row is one training observation. With this fixed small column count, least-squares work grows approximately linearly with training length. Filtering two states also takes linear time and stores a two-column probability array over the sequence. The reference router passes five contract tests; these tests exercise declared behavior rather than proving economic assumptions.

Training states and nominal filter parameters are oracle inputs. Results therefore isolate specified failure modes and do not establish performance with learned regimes. The evidence router and statistical diagnostic are separate modules: the routing weights from evidence scores are not the filter probabilities used in Table 1. No end-to-end experiment combines language-model proposals, graph identification, estimation, regime learning, and trade execution.

The synthetic graph is deliberately simple, with a known treatment effect and independent disturbances. Real economic mechanisms may involve simultaneous feedback, unobserved interventions, revised data vintages, weak support, and multiple plausible graphs. Confidence in a predicted regime is not a calibrated probability that a causal assumption is true. The declaration of identification can itself be wrong; the current router cannot discover that fact.

8 Conclusion

CMTF makes the boundary between a proposed mechanism and an eligible decision explicit. Its implemented contribution is a small, inspectable contract for declared evidence; its empirical contribution is a reproducible diagnostic showing how confounding and regime-observation error affect coefficient routing. Under a valid observation model, soft routing improves coefficient recovery. Under hidden confounding, even oracle routing remains biased. Under reversed state semantics, filtered routing becomes worse than pooled adjustment. These outcomes support explicit identification status, provenance, and failure handling, while leaving live economic effectiveness and full neuro-symbolic integration open to separate evaluation.

AI Use Statement

ChatGPT/Codex was used for conceptual development, mathematical exposition, research-method design, implementation and testing, synthetic-data generation code, data cleaning and auditing, qualitative analysis, and interpretation. It also assisted literature discovery, manuscript drafting and editing, figure preparation, reference formatting, and artifact packaging. No human-subject translation or participant study was performed. Automated checks include reproducibility runs, numerical comparisons, contract tests, and manuscript build checks; these do not replace author review or external replication. The author is responsible for the final claims, attribution, and artifacts. The generated synthetic observations and hypothetical policies are explicitly distinguished from operational evidence throughout the manuscript.

Data, code, and scope

The synthetic study and public aggregate results accompany this paper. No provider keys, private trading logs, account identifiers, or original model prompts are included in the public release. The public code does not execute trades or invoke paid model services. Operational results, where discussed, are descriptive source audits with restricted raw-data availability. This paper does not report investment advice, verified risk-adjusted profitability, or a causal effect in an actual market.

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A Algorithm and code mapping

In `synthetic_study.py`, `state` stores the integer regime sequence, `c/x/y/m` store the common cause, treatment, outcome, and regime observation, and `q` stores a matrix with one row per time and two state-probability columns. `A`, `MEANS`, and `SIGMA` implement the transition matrix, emission means, and observation standard deviation. `beta` is the two-element vector of fitted state-specific slopes. Matrix multiplication `q @ beta` takes the probability-weighted sum for each row and returns a vector of coefficient predictions.

The generator first samples a state sequence, then independent disturbances, then constructs treatment, outcome, and observation arrays. The filter processes observations sequentially. Regression uses only the training segment and oracle training states. Predictions are evaluated on test coefficients supplied by the simulator. This ordering isolates observation-model failure and avoids fitting regression coefficients on test outcomes.

The router processes a list of immutable evidence records. It first checks unique identifiers, then records a reason for each rejected item. Valid logits enter a stable exponential normalization. An empty eligible list returns abstention. The output dictionary contains status, weights, and rejection reasons. It contains no broker order and does not execute any action.

B Paired differences and numerical scope

The mean hard-minus-soft MSE differences are 0.129911 (Monte Carlo standard error 0.002849), 0.131984 (0.005207), 0.610536 (0.003371), and 0.032484 (0.000783) for observed confounding, hidden confounding, emission reversal, and inactive second regime, respectively. Each difference is first computed within the same replicate. Its standard deviation across 200 replicates is divided by $\sqrt{200}$ to obtain the reported standard error. This avoids treating correlated methods as independent observations.

The conditional-mean derivation in the main text assumes exact coefficients and correct posterior probabilities. The experiment uses estimated coefficients and, deliberately in one scenario, wrong emission parameters. The mathematical identity is an explanation of the nominal behavior, not a guarantee for the implemented filter under misspecification.

C Execution and artifact availability

Run `python reproduce.py` at the public package root. The diagnostic outputs include raw replicate MSE and mean absolute error, summary means and standard errors, state-recovery diagnostics, paired comparisons, and an environment manifest. Mean absolute error averages the absolute rather than squared coefficient deviation and is provided as a secondary diagnostic. No raw financial or user data are required for this paper’s numerical tables.

The implementation uses CPU-based NumPy and pandas; no GPU or provider endpoint is invoked. The archived environment specifies Python 3.12.14, NumPy 2.3.5, pandas 2.2.3, SciPy 1.17.0, and Matplotlib 3.10.8. The seed schedule and nominal parameters are fixed in the supplied script. Five router contract tests cover implemented behavior. Full statistical results consist of 800 generated datasets and 4,000 method rows.

D Relationship to the companion study

The companion market study uses an operational audit, exported directional responses, and a separate price-feedback simulator. It does not evaluate this evidence router as a live causal controller. Conversely, the present study does not reuse market returns as evidence of causal identification. The two papers share motivation and software packaging but have distinct outcome measures, data-generating processes, and empirical claims. The original conceptual note is cited as prior work rather than described as an independent evaluation.